

Overnight report, 2026-09-13/14: verbose levels, STD strength matching, and seven manuscript improvements

Session 90c5825b on R5611351, branch main, commits f5c5cac .. see §9. Plan: ~/.claude/plans/dreamy-dazzling-ladybug.md. Everything ran through the MATLAB MCP; cClear cClasses was called several times after classdef edits, which WIPED THE BASE WORKSPACE of the live MATLAB session (the variables cfg, cfg3, results, results3, run_dir, run_dir3, zz_*, r_* left in it are mine).

0. The headline: the scaled matching makes the multiple-timescale network chaotic

TR chose (2026-09-13) to match STD strength by a **multiplicative route scale** $s = 1 + r_{\text{ref}}/\rho = 3$ on the two-timescale routes, with the usage-matched (τ_{rel}) variant as a control. Both were built, tested and run at fast (data/stdscaled_fast, data/stdusage_fast; figures in figs/stdscaled_fast, figs/stdusage_fast). The scale variant does what the plan's context warned it would, and more:

	unmatched (data/topk_med, medium, 15 reps)	scaled, $s = 3$ (data/stdscaled_fast, fast, 3 reps)	usage, $\rho_u = 0.342$ (data/stdusage_fast, fast, 3 reps)
lambda_1 near default: none / single / multiple example run $n = 500$, 20 s, seeds [1 2]	+3.47 / +0.35 / -0.112 (105 of 105 negative) +3.35 / +0.43 / -0.11	+3.65 / +0.61 / +2.24 (0 of 21 negative) +3.77 / +0.43 / +1.99	+3.65 / +0.61 / +1.01 (0 of 21 negative) +3.77 / +0.43 / +0.90
tau sweep, multiple-timescale, slowest tau 1 -> 30 s	-0.112 -> -0.043, every rep negative	+2.27 -> +2.27, every rep positive	+0.86 -> +1.14, every rep positive, no trend
memory capacity, total MC: none / single / multiple K needed by top-K for the multiple-timescale condition	0.103 / 0.259 / 0.590 (15 trials, $p = 6e-5$) 15	0.111 / 0.243 / 0.103 (5 trials) 60, D_KY resolved in 0-50% of jobs	0.111 / 0.243 / 0.146 (5 trials) 15-60 (D_KY resolved in 39-100% of jobs)
1-D sweeps: multiple-timescale is the least stable condition on	none of 7	f_E, mu_EE, mu_EI, mu_II (4 of 7)	n, mu_EI, mu_IE, mu_II (above the single-timescale curve on 6 of 7 sweeps; below it only on mu_EE)

With the 3x undepressed recurrent gain the multiple-timescale network is strongly chaotic and its fading memory collapses to the no-adaptation level; the paper's central computational result does not survive the scale direction. The usage control (R3) is less extreme but points the same way: $\lambda_1 +1.0$ near default, +0.9 in the example run, +0.65 to +1.14 across the tau sweep with no trend, memory capacity 0.146. **At equal steady-state depression strength at r_{ref} , the timescale count alone does not produce the stable regime: the unmatched multiple-timescale advantage leaned on the squared (stronger) depression.** That is the honest answer to Codex sec. 1's confound question, and it is what the manuscript must say.

Because both matchings WEAKEN the multiple-timescale network's depression, a third direction was built and launched at 03:15 (sec. 8): keep the multiple-timescale network exactly as published and STRENGTHEN the single-timescale route to the dual's steady state ($\tau_{\text{rel}} 0.25 \rightarrow 0.0625$, ..._dualStdSingleMatched_3cond_mu8p25). If a single-timescale STD of equal strength also gives $\lambda_1 \sim -0.1$ and MC ~ 0.6 , the paper's claim is about depression strength, not timescale count; if it does not, the timescale structure matters at that strength. paper_config still points at the SCALED preset because that was the instruction; **it should not stay there.** Nothing has been run at medium on any matched variant except the usage run launched tonight (data/stdusage_med).

The matching itself is exact at r_{ref} for both variants (fig_STD_strength_matching): $\theta_{\text{ss}}(r_{\text{ref}}) = 0.0833$ for single, scaled and usage; the largest |log ratio| to the single-timescale route over the occupied rate band [0.199, 0.254]

is 1.11 unmatched, 0.146 scaled, 0.034 usage. Usage is also the better match away from `r_ref`. Caveat on record: per-neuron time-averaged rates are strongly bimodal (near 0 or saturated), so “the operating point” is the population mean rate, not a typical neuron’s rate.

1. `r_ref`

`r_ref` = 0.25 = the median mean firing rate of the multiple-timescale condition at the default point of the five well-defined 1-D sweeps of `data/topk_med` (`f_E` 0.236, `level_of_chaos` 0.223, `mu_EE` 0.239, `mu_IE` 0.251, `n` 0.231; with the two negative-mean sweeps resolved correctly the pooled 105 reps give 0.230 [p5 0.199, p95 0.254]), rounded to 0.05. Single-timescale 0.269, no adaptation 0.483. The joint 64-point sample, far from the default, has 0.154 / 0.159 / 0.250. Full tables and the replacement manuscript Methods paragraph: `docs/notes/STD_strength_matching_2026-09-13.md` (+pdf).

2. What was built (all committed, all tested; §9 lists the commits)

- **Verbose** (`f5c5cac`): one setting, `'verbose'` | `'minimal'` | `'near-none'`, `minimal` the default, from `paper_config` through `ctx`, every stage, `ParamSpaceAnalysis2` and both model classes into the `parfor` workers; the `evalc` wrappers in the stages are gone; warnings never gated. A serial 2 x 2 sweep prints three lines. `test_verbose_levels`.
- **STD strength matching** (`d942c51`): `synapse_config.<pre>.<post>.scale` (`params.W` only; `obj.W` untouched; dynamics, both Jacobians and `jacobian_times` consistent by construction; the ESN readout check compares it); presets `..._dualStdScaled_3cond_mu8p25` (primary) and `..._dualStdUsage_3cond_mu8p25` (control); golden fixture regenerated (16 presets); `fig_STD_strength_matching` (`in_paper`); `std-Scaled_fast_config/_run`, `stdUsage_fast_config/_run`; `Equations_stability_paper.md` subsection, `MTS_STD_notes.md` addendum, `CLAUDE.md`. `test_route_scale`. The manuscript was NOT edited; its “depression is deliberately not normalized” sentence and the replacement are in the note.
- **Tau sweep on both cell types** (`b317923`): `tau_a_EI` alias; the stage picks `tau_a_EI` when the condition’s E and I ladders are identical; `tau_levels.mat/.mat` manifest proves it (R2: equal E and I ladders at all 7 levels). `test_tau_a_EI_alias`. The sweep folder is now named by its axis.
- **Numerics gate** (`c0a2a22`): check J (analytic Jacobian vs central finite differences on the reduced net, kink rows excluded); acceptance thresholds fixed in code on 2026-09-14 before any run, saved in `settings.acceptance`; verdict in the `.mat` and `numerics_verdict.md`; `'jacobian'` figure variant. `test_numerics_verification_stage` (~13 min).
- **Imbalance eig-heatmap examples** (`c0a2a22`): `run_eig_heatmap cfg.examples` (`mu_EE` x 0.5 / 1 / 1.5, shared seeds, per example and condition eigenvalues, top-K `lambda_1`, mean rate and `B_E`); `fig_eig_heatmap_imbalance` (`in_paper`); legacy variables unchanged. `test_eig_heatmap_stage`.
- **Figures** (`07dc9a6`): `fig_example_timeseries` three-column composite with the local rate and the accumulating finite-time `lambda_1` (`plot_local_and_finite_lle`, `SRNNCellTypePairs.routes_identical`); `fig_local_vs_finite_lle`; `fig_lle_vs_rate`; IQR bands and data-driven limits on `fig_sensitivity_medians`; the unclipped density-strip `fig_sfa_EOC_allStd`; E:I weights title overlap.
- **Memory capacity** (this commit): `fig_memory_capacity` redrawn (data-driven cumulative axis, bootstrap bands, paired horizons with the sign-flip `p` and `d_z` read from the saved stats, a table); `run_memory_capacity` writes `<run>_provenance.md` (commit, preset, mode, seeds, trials, protocol, stats, paths). The 30-trial production run: see §8.

3. Numerics verdict (fast, scaled preset; `data/stdscaled_fast/numerics_verification/numerics_ver`

check	none	single	multiple	note
A noise-free reshoot order slope ≥ 1	2.01 PASS	2.01 PASS	2.01 PASS	Ralston order 2 despite the kinks
B noisy strong-order slope ≥ 1	2.01 PASS	1.98 PASS	1.99 PASS	ratio 4x per halving

check	none	single	multiple	note
L median $ \lambda_1 - \text{ode45} \leq 0.05$	0.108 FAIL	0.169 FAIL	0.722 FAIL	10-s windows, 2 seeds: finite-time scatter
C Benettin - QR ≤ 0.05 , $ \text{topK} - \text{QR} \leq 0.02$	0.037 / 0.006 PASS	0.308 / 0.016 FAIL	0.166 / 0.058 FAIL	reduced net n = 30, T 12 s
J rel Frobenius, eigenvalue error $\leq 1e-6$	1e-10 / 7e-9 PASS	5e-11 / 5e-10 PASS	6e-11 / 5e-10 PASS	0 kink rows excluded

The gate works as designed: the convergence and Jacobian claims pass; the finite-time LLE agreement claims cannot be judged on fast windows and must be judged at medium (25 seeds, 20-s windows) or production. The manuscript's Jacobian claim now has a saved artifact.

4. Tau sweep on both cell types

data/topk_med (unmatched, tau_a_E only, 11 levels x 25 reps): every rep negative, λ_1 -0.112 at 1 s rising monotonically to -0.043 at 30 s ($-1/\tau = -0.033$): the slowest SFA rung sets the exponent in the stable regime, as the sweep's header predicts. The old figure's "half the distribution positive" described a Benettin-era run. R2 (scaled, tau_a_EI): every rep positive, +2.3 at every level – the network is chaotic and the SFA timescale no longer sets λ_1 . R3: +0.86 $\rightarrow$ +1.14, every rep positive, no trend_TEXT.

5. Local vs finite-time LLE (fig_local_vs_finite_lle) and LLE vs rate (fig_lle_vs_rate)

data/topk_med, near-default set (7 sweeps at the level nearest the default, 105 networks per condition):

condition	λ_1 median [IQR]	networks with $\lambda_1 < 0$	local rate > 0, share of time	median frac_local_positive	mean excursion (s)
no adaptation	+3.47 [+1.81, +4.33]	4 / 105	83%	0.91	0.70
single-timescale	+0.35 [-0.06, +0.86]	33 / 105	53%	0.54	0.26
multiple-timescale	-0.112 [-0.113, -0.110]	105 / 105	3%	0.02	0.12

So "transient expansion within long-interval stability" (Codex §3's stronger claim) describes the SINGLE-timescale regime near the boundary (locally expanding half the time, λ_1 straddling zero), not the multiple-timescale one, which is quietly stable (locally expanding 3% of the time, λ_1 pinned at -0.11 by the 10-s SFA rung). Keep the stronger idea in the Discussion.

LLE vs mean rate over 1219 networks (joint sample + 1-D sweeps), Spearman rho [bootstrap 95% CI]: no adaptation -0.44 [-0.50, -0.38] (inverted U: quiet +0.3, mid +3.1, saturated -5.7), single -0.31 [-0.36, -0.25], multiple +0.11 [+0.05, +0.16] (flat at -0.11 across the whole rate range). Rate and stability interact through the operating point and are not interchangeable; in the multiple-timescale regime the rate barely predicts λ_1 at all.

6. Imbalance eig-heatmap examples (R2, scaled preset, fast: reference at n = 500, imbalanced at n = 250)

example	condition	lambda_1	mean rate	B_E
mu_EE x 0.5 (inhibition-dominant)	none / single / multiple	+1.98 / -0.19 / +4.89	0.06 / 0.05 / 0.05	0.42
reference	none / single / multiple	+3.94 / +0.63 / +1.95	0.43 / 0.25 / 0.30	0.50
mu_EE x 1.5 (excitation-dominant)	none / single / multiple	-4.23 / -0.53 / +1.19	0.87 / 0.71 / 0.35	0.55

The sheet (figs/stdscaled_fast/fig_eig_heatmap_imbalance) reads well: the no-adaptation network goes from chaotic to saturated-and-silent as excitation grows, single-timescale stays near zero, and the scaled multiple-timescale network is chaotic in every column (its occupancy spills past $Re = 0$ in all three rows). Rerun on the chosen preset at medium before it goes in the paper.

7. Transient gain on the scaled preset (fast, for the record)

Operating-point table: lambda_1 +4.0 / +0.44 / +1.95; alpha(J_xx) at state +8.7 / +7.2 / +6.7; alpha(J) +8.7 / +2.6 / +4.0. Adaptation's feedback still takes the frozen rate from +6.7 to +4.0 in the multiple-timescale regime, but the trajectory itself is at +1.95: the scale has moved the operating point, not removed the mechanism.

8. Memory capacity

Audit of data/topk_med/memory_capacity/MC_sample_hold_20260912_232101_trials15_* (unmatched preset, medium): 15 of 15 paired trials complete for all three conditions, all summaries finite; total MC 0.103 [0.093, 0.114] / 0.259 [0.218, 0.313] / 0.590 [0.515, 0.664]; horizon 0.00 / 0.12 / 0.52 s; exact sign-flip over 32768 patterns: none vs multiple $p = 6.1e-5$, $d_z = -3.32$; single vs multiple $p = 1.2e-4$, $d_z = -2.04$. fig_memory_capacity resolves exactly that file for run_dir = data/topk_med. Horizons are quantised to 0.3-s holds, so panel (c) is inherently discrete.

R2 (scaled, fast, 5 trials): 0.111 / 0.243 / 0.103, single vs multiple $p = 0.0625$ (the exact floor at 5 trials) with $d_z = +1.66$ in the WRONG direction. R3 (usage): 0.111 / 0.243 / **0.146** (5 trials)_TEXT.

Unattended chain launched at 03:15 in one MATLAB call (diaries in data/): 1. stdSingleMatched_fast_run -> data/stdsinglematched_fast, figs/stdsinglematched_fast (log data/stdsinglematched_fast_log.txt; ~1.7 h); 2. stdUsage_med_run -> data/stdusage_med, figs/stdusage_med (medium: 15 reps, 11 levels, 15 MC trials; equal footing with data/topk_med; ~3 h); 3. run_memory_capacity at PRODUCTION (30 paired trials) on the UNMATCHED paper preset -> data/topk_med_mc_production (log data/mc_production_log.txt). This replaces the plan's production MC on the scaled preset, whose fast result is already decisive. Plot it with fig_memory_capacity('run_dir', 'data/topk_med_mc_production', 'out_dir', ...) (the runner writes into <output_dir>/memory_capacity). If MATLAB is still busy in the morning, the diaries say which job it is on; run_all_paper_analyses refuses a non-empty run directory, so a crashed job can be restarted after deleting its folder.

9. Commits, figures, what was left undone

Commits tonight, in order: f5c5cac verbose; d942c51 STD matching; b317923 tau_a_EI; c0a2a22 numerics gate + imbalance examples; 07dc9a6 figures; 9e39cb3 config handles; a1bb18b docs; eb8e56d make_all_paper_figures verbose order (this bug cost R2 one figure pass: the level was read at line 66 and defined at line 119); 35f70bc memory capacity figure + provenance; 336532f the third matching direction (dualStdSingleMatched) and the medium usage config; the final commit carries this note and the fig_STD_strength_matching title fix.

Figures: figs/stdscaled_fast/<entry>/ (31 of 31 entries succeeded, 17 in-paper, manifest.md there); figs/stdusage_fast/ (31 of 31, 17 in-paper). Scratch renders from the unmatched medium data are in figs/scratch/.

Wrong turns worth knowing: the Bash tool mangles backslashes in heredocs, so every multi-line edit went through perl scripts written with the Write tool (two figure registrations lost their @ handles that way, 9e39cb3); test scripts run in the base workspace and clobber loop variables, so tests were run one per call; a (1, :) char argument rejects the logical the old callers pass, so the verbose arguments are unconstrained and validated by verbose_name; MC's capture_git_provenance writes a working_changes.patch when the tree is dirty, which it was during the runs (the figure/provenance edits were uncommitted while R2 ran).

Not done: nothing from the plan was skipped. Raised only: the E:I weights sheet's colour scheme and the sensitivity sheets still show the no-adaptation curve dominating the y range (the adapted curves are readable but compressed; a symmetric-log axis would help); fig_local_vs_finite_lle's row-2 x labels overlap between columns; plot_memory_capacity_combined still has the fixed [0 10.9] cumulative axis for direct callers; test_numerics_verification_stage takes 13 min because the full-size L check has no size override.

10. What TR should do first

1. Read sec. 0. Both weakening matchings (scale, usage) turn the multiple-timescale network chaotic and remove its memory advantage. Look at data/stdsinglematched_fast (sec. 8 chain, job 1): does a single-timescale STD of the dual's strength reproduce $\lambda_1 \sim -0.1$ and MC ~ 0.6 ? Read figs/stdsinglematched_fast/fig_sensitivity_medians, fig_memory_capacity and fig_local_vs_finite_lle tables.
2. Decide what the paper claims. Options: (a) the timescale-count claim, supported only if single-matched does NOT reproduce the stable regime; (b) a combined "multiple timescales AND stronger depression" architecture claim, which the unmatched comparison supports and which the Methods paragraph in STD_strength_matching_2026-09-13.md sec. 4 must then be rewritten to say; (c) match at a different quantity (e.g. the occupied mean of theta rather than theta at one rate).
3. Move paper_config off the scaled preset (one line) to whichever preset the decision names; the medium usage run (data/stdusage_med) and the 30-trial production MC on the unmatched preset (data/topk_med_mc_production) will be waiting either way.